

ApexPlanet Data Analytics Internship

Task 5 Final Report

E-Commerce Sales Analytics using Python, SQL & Power BI

Executive Summary

This report presents the analysis of the Superstore Sales dataset using Python, SQL, statistical techniques, and Power BI. The workflow included data cleaning, exploratory data analysis, SQL-based business queries, statistical summaries, and dashboard development to generate actionable business insights.

Problem Statement & Objectives

The objective was to analyze historical sales data to identify revenue drivers, profitable products, customer segments, regional performance, and operational opportunities that support data-driven decision making.

Data Source & Methodology

Dataset: Superstore Sales (9,994 records, 21 columns). Workflow: Data Cleaning → EDA → SQL Analysis → Statistical Analysis → Dashboard Development → Business Insights → Recommendations.

Key Findings

- Technology was the highest-profit category.
- West region achieved the highest sales.
- Consumer segment generated the greatest profit.
- Standard Class dominated shipping.
- California produced the highest state profit.
- Accessories and Phones were the most profitable sub-categories.
- COD was the leading payment mode.

Statistical Analysis

Data quality assessment showed no missing values or duplicate records before outlier treatment. Data types were standardized and IQR-based outlier removal produced a cleaned dataset of 6,764 records for further analysis.

Dashboard Summary

KPIs: Total Sales = 1.6M, Total Orders = 22K, Total Profit = 175K, Average Shipping Days = 4. Dashboard visuals include monthly sales and profit trends, category and regional performance, payment modes, shipping modes, and state-wise profit.

Recommendations

Increase focus on high-profit technology products, optimize discount strategies, strengthen marketing in high-performing regions, improve customer retention, and monitor KPIs through interactive dashboards.

Future Scope & Limitations

Future enhancements include demand forecasting, machine learning, customer segmentation, and inventory optimization. Limitations include historical data only and absence of external market factors.